

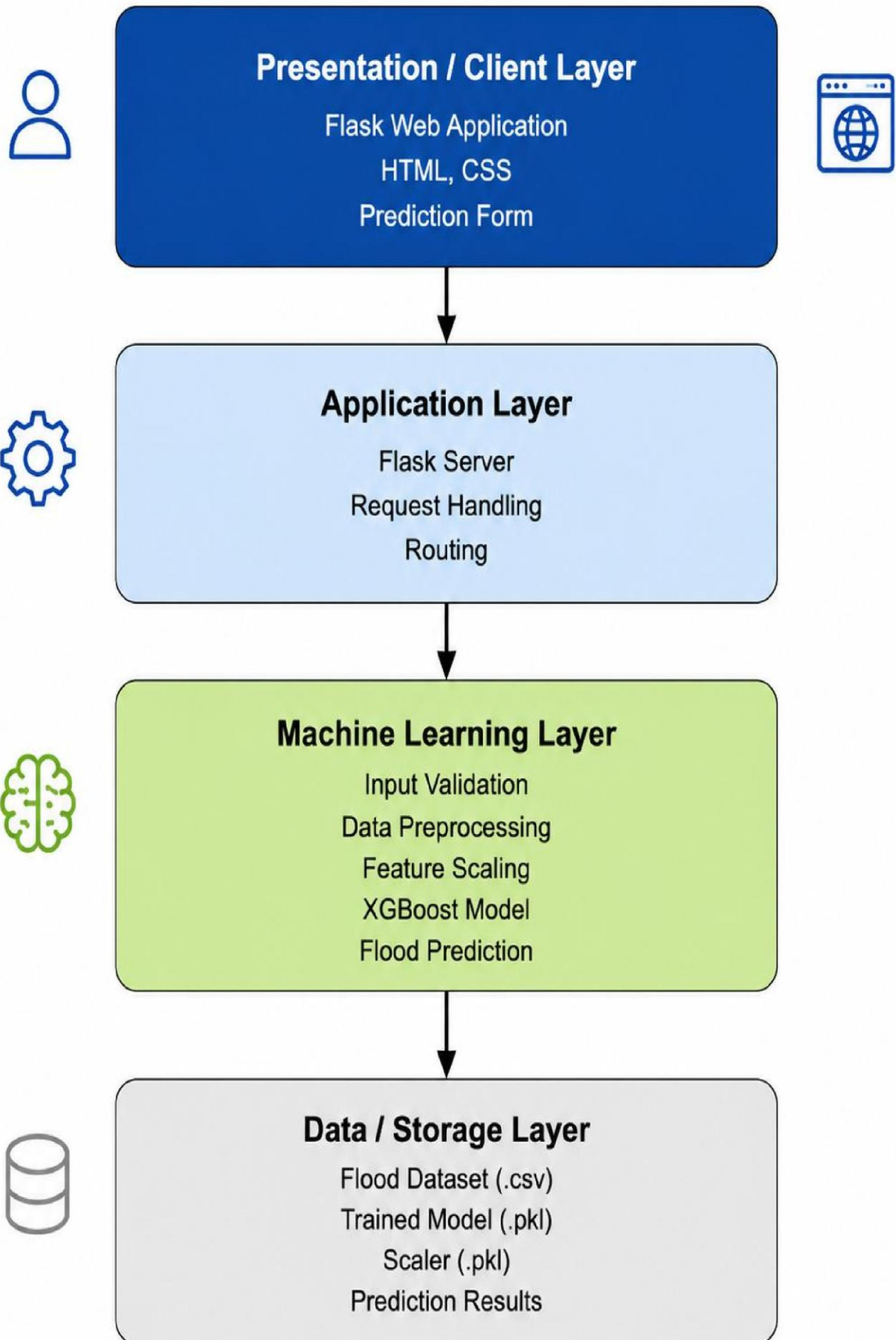
Solution Architecture

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	Provides a user-friendly web interface where users enter weather-related data and view the flood prediction results.	Flask, HTML, CSS
Application Layer	Receives user requests, validates the input, processes the data, and communicates with the machine learning model.	Flask, Python
<i>Machine Learning Layer</i>	Preprocesses the input data, performs feature scaling, loads the trained model, and predicts flood risk.	<i>Scikit-learn, XGBoost, Pandas, NumPy</i>
Data / Storage Layer	<i>Stores the flood dataset, trained machine learning model, and scaler used for prediction.</i>	<i>CSV Dataset, Joblib (.pkl files)</i>